Operating Systems

UE23CS242B

4th Semester, Academic Year 2023

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TITLE:

USING MUTEX LOCKS, WRITE A PROGRAM THAT CREATES MULTIPLE THREADS AND PERFORMS A PARALLEL SUM OF AN ARRAY OF INTEGERS. EACH THREAD SHOULD WORK ON A PORTION OF THE ARRAY AND UPDATE A GLOBAL VARIABLE THAT STORES THE SUM. THE PROGRAM SHOULD USE MUTEX LOCKS TO PROTECT THE ACCESS TO THE GLOBAL VARIABLE AND AVOID RACE CONDITIONS.

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/shm.h>

#include <string.h>
```

```
#include <pthread.h>

pthread_t tid[5]; //Using 5 threads
pthread_mutex_t lock;
int sum = 0;

//function to sum
void *add(void *arg)
{
    int *array = (int *)arg;

    pthread_mutex_lock(&lock);

    for(int i = 0; i < 2; i++){
        sum += array[i];
    }

    pthread_mutex_unlock(&lock);
    return NULL;
```

```
}

int main(void)
{
    int array[10] = {1,2,3,4,5,6,7,8,9,10};

    if (pthread_mutex_init(&lock, NULL) != 0)
    {
      printf("Mutex init has failed\n");
      return 1;
    }

    for(int i = 0; i < 5; i++){
        pthread_create(&tid[i], NULL, add, (void *)&array[i*2]);
    }

    for(int i = 0; i < 5; i++){
        pthread_join(tid[i], NULL);
    }
```

```
    printf("Sum of array is %d\n", sum);
    pthread_mutex_destroy(&lock);
    return 0;
}
```

Output:

```
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/My Documents/PES/Sem 4/OS/codes# gcc assignment2.c
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/My Documents/PES/Sem 4/OS/codes# ./a.out
Sum of array is 55
root@LAPTOP-10GABJ15:/mnt/c/Users/Roshini/My Documents/PES/Sem 4/OS/codes#
```